Integration — Lesson 27 Test: FTC Part 2, Derivatives of Accumulation Functions

10 questions · show your work in the box · no calculator needed · justify where asked

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Find $\frac{d}{dx} \int_3^x t^4 \, dt$. 1 pt

2. Find $\frac{d}{dx} \int_0^x \sqrt{1 + t^2} \, dt$. 1 pt

3. Let $g(x) = \int_5^x \cos t \, dt$. Find $g(5)$ and $g'(x)$. (One of them needs no computing.) 1 pt

4. A student defines $g(x) = \int_0^x x^3 \, dx$. Explain what is wrong with this notation and write it correctly. Why must the inside use a different letter, like t? 1 pt

5. Find $d/dx \int_0^{x^2} \cos t \, dt$. 1 pt

6. Find $d/dx \int_2^{x^3} \sqrt{1 + t} \, dt$. 1 pt

7. Find $d/dx \int_x^4 (t^3 + 1) \, dt$. (Watch where the x is!) 1 pt

8. Let $g(x) = \int_0^x (t^2 - 9) \, dt$ for $x \geq 0$. Find $g(0)$, $g'(x)$, and $g'(4)$, and state with justification the interval on which g is decreasing. 1 pt

9. Let $g(x) = \int_1^x 3t^2\, dt$. Write the equation of the tangent line to the graph of g at x = 2. (Find g(2) with FTC Part 1, and get the slope from FTC Part 2.)

1 pt

10. A student wrote: d/dx ∫0^(3x) sin t dt = sin(3x). Find the error, explain it in one sentence, and give the correct answer.

1 pt